

Gayathre V

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Professional Summary

Final year M.Sc. Data Science student specialising in NLP, computer vision, and deep learning. Experienced in building transformer-based pipelines, RAG systems, and CNN-based medical image classifiers, with peer-reviewed research in toxicity detection and Alzheimer's MRI classification.

Education

Integrated M.Sc. in Data Science

Amrita Vishwa Vidyapeetham, Coimbatore

CGPA: 7.79 / 10.00

Skills Summary

Languages: Python, SQL, R | **Databases:** MySQL, SQLite, MongoDB, Cassandra, Redis

ML & Data Science: Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, SciPy, SHAP, MLflow

Deep Learning & NLP: PyTorch, TensorFlow, Hugging Face, Transformers, LSTM, Vision Transformers

LLMs & RAG: LangChain, LangGraph, ChromaDB, Sentence Transformers, Prompt Engineering, Vector Search

Backend & Frontend: FastAPI, Flask, React, Next.js, Streamlit, REST APIs | **Tools:** Docker, Git, Azure, Tableau

Projects

LegalLens — AI Contract Intelligence System

 GitHub

Python, LangChain, ChromaDB, Sentence Transformers, Streamlit

- Built a multi-agent RAG system for contract document understanding, extracting risks, obligations, and key clauses using semantic search over dense vector embeddings.
- Designed an NLP pipeline combining chunking strategies, embedding models, and LLM reasoning to enable natural language querying over legal text.

SentimentPulse — Live NLP News Sentiment Dashboard

 GitHub

Python, HuggingFace Transformers (DistilBERT), TF-IDF, KMeans, Streamlit, Plotly

- Built a web scraping and NLP pipeline that collects live headlines from multiple news sources and classifies sentiment using transformer-based models.
- Designed an interactive dashboard to visualise sentiment trends across topics and publications in real time, enabling media tone analysis at scale.

AI Research Assistant — Multimodal Document QA System

 GitHub

Python, LangChain, RAG, PDF Parsing, Generative AI, Streamlit

- Built a multimodal RAG system that parses and indexes PDF research papers, enabling natural language Q&A over scientific documents with context-grounded responses.
- Combined document layout parsing, semantic chunking, and retrieval-augmented generation to handle dense academic text including figures and structured tables.

Research & Publications

Bias-Aware Toxicity Detection in Social Networks

Accepted, IECCT 2026

- Built a bias-aware NLP pipeline combining classical ML, deep learning, and transformer models to improve fairness in toxicity classification on the Jigsaw dataset.
- Reduced false positives on identity-related text across sensitive demographic groups using BERT-family models with group-level fairness metrics.

Alzheimer's Disease Classification via Deep Learning Ensembles

Communicated, Informatics in Medicine Unlocked

- Developed an ensemble deep learning model for MRI-based Alzheimer's stage classification with Grad-CAM explainability, validating focus on clinically relevant brain regions.
- Applied image preprocessing and augmentation pipelines to improve model generalisation on multi-class neuroimaging datasets.

Certifications

Google Data Analytics Certificate — Google / Coursera

Data cleaning, visualisation, and analysis using SQL, Excel, and Tableau.

Introduction to Machine Learning — NPTEL

Supervised and unsupervised learning, model evaluation, and regularisation with Python.

Business Analytics and Text Mining Modelling using Python — NPTEL

Python fundamentals to advanced scientific computing with NumPy, SciPy, and Pandas.